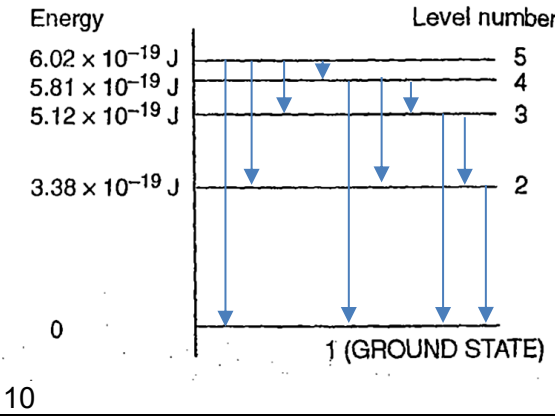
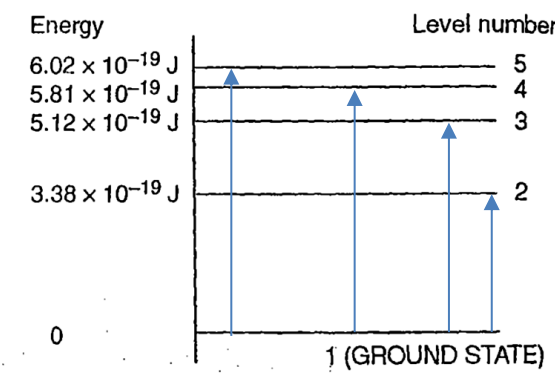
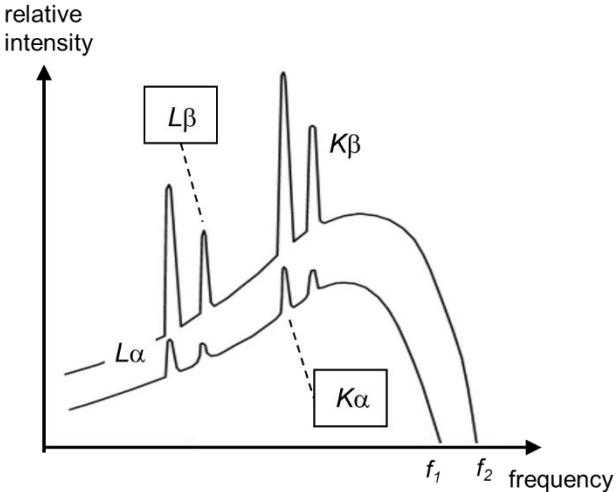


Questions			Answers	Marks
7	(a)	(i)	Longest wavelength $\rightarrow$ lowest energy photon Therefore emission transition must be between two closest levels, <u>5 to 4</u> Do not accept 4 to 5	[B1]
		(ii)	 Energy Level number 6.02 $\times$ 10⁻¹⁹ J 5.81 $\times$ 10⁻¹⁹ J 5.12 $\times$ 10⁻¹⁹ J 3.38 $\times$ 10⁻¹⁹ J 0 1 (GROUND STATE) 10	[B1]
		(iii)	Converting 3.6 eV into joules, 5.76 $\times$ 10⁻¹⁹ J is available from the bombarding electrons 5.76 is larger than 5.12, but smaller than 5.81, so maximum absorption of energy will be up to 5.12 Therefore, observed transitions will be between Level 3 and Level 1, i.e. 3	[M1] [M1] [A1]
	(b)		 Energy Level number 6.02 $\times$ 10⁻¹⁹ J 5.81 $\times$ 10⁻¹⁹ J 5.12 $\times$ 10⁻¹⁹ J 3.38 $\times$ 10⁻¹⁹ J 0 1 (GROUND STATE) For transitions to be visible to human eye, consider extreme values of 400 nm and 700 nm 400 nm: Using $E = hc/\lambda = (6.63 \times 10^{-34})(3.00 \times 10^8)/(400 \times 10^{-9}) = 4.9725 \times 10^{-19}$ J 700 nm: $(6.63 \times 10^{-34})(3.00 \times 10^8)/(700 \times 10^{-9}) = 2.8414 \times 10^{-19}$ J From energy levels in Fig. 7.1, only 1 (one) visible line (from Level 1 to Level 2)	[M1] [A1]

(c)			[B1]
(d)		Accelerated <u>incident electron 'collides' with an electron from the innermost shell</u> (step 1) <u>kicking it out of the K-shell</u> (step 2). The atom is excited due to the vacancy in the K-shell. <u>An electron from the M-shells transits to the vacancy in the K-shell</u> (step 3), emitting a photon in the x-ray frequency, <u>corresponding to the peaks</u>	[B1] [B1]
		Alternate: The <u>high energy incident electrons colliding with the metal target</u> give the <u>innermost shell electrons sufficient energy to be removed</u> from the metal or transited to higher energy levels. The <u>peaks correspond to the characteristic X-rays that are emitted</u> when <u>M-shell electrons transits to the lower energy shells</u>.	[B1] [B1]
(e)		With <u>higher accelerating voltage</u> there will be <u>greater number of electrons that can be removed and accelerated from the filament</u>. Also, generally <u>electrons with higher kinetic energy undergoing deceleration</u> should be able to produce more photons. Hence the higher intensity of the broad spectra with higher accelerating voltage	[B1] [B1]
(f)		The <u>peak frequencies are dependent on the energy transition of the other higher energy shell electrons to the lower energy shells</u>, in which discrete quanta of energy are emitted, <u>depending on energy level structure of the metal</u>. This is independent of applied voltage.	[B1] [B1]

